

NovaBank

Banking Transaction Analytics Pipeline

End-to-End Data Engineering + Analytics + Anomaly Monitoring
Portfolio Project Report

This project uses synthetic banking data generated solely for demonstrating data engineering, analytics, and transaction-monitoring capabilities. It does not represent real customer or banking data.

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GitHub	https://github.com/alirazach476/banking-transaction-analytics
Live demo	https://banking-transaction-analytics.vercel.app
BI dashboard	https://banking-transaction-analytics.vercel.app/dashboard
Report date	07 October 2026
Stack	Python, PostgreSQL, dbt, Airflow, scikit-learn, Docker, Vercel

1. Project Overview

NovaBank is a fictional retail bank requiring a centralized analytics platform for transaction monitoring, customer/account behavior, channel and branch performance, operational failures, and a Transaction Anomaly Detection Prototype. This portfolio delivers a runnable, tested, dockerized pipeline with dimensional modeling, reconciliation, and a live BI dashboard.

Business goals

- Single source of truth for transaction analytics
- Reliable KPIs for finance, operations, and channel performance
- Controlled anomaly review queue (not production fraud decisioning)
- Interview-ready demonstration of DE / AE / analyst skills

2. Architecture

Pipeline flow: Synthetic source systems → Ingestion → Raw layer → Data quality → PostgreSQL → dbt (staging / intermediate / warehouse / analytics) → Anomaly detection → BI dashboard / SQL analytics / Airflow orchestration.

Schemas: raw, staging, intermediate, warehouse, analytics, audit, monitoring.

Fact grain: One row in fact_transactions represents one financial transaction event.

3. Dataset Statistics (Verified Demo Run)

Configurable volumes via .env. Figures below are from the verified end-to-end run.

Entity	Count / Value
Customers	5,000
Accounts	7,500
Transactions (fact)	101,600
Transfers	15,000
ATM transactions	20,000
Card transactions	40,000
Branches / Merchants / Cards	50 / 1,000 / 10,000
Total transaction value	\$81,172,698.21
Average transaction	\$799.18
Anomalies requiring review	8,645
Daily mart rows	1,095

Full portfolio targets (configurable): 100k customers, 150k accounts, 2M+ transactions.

4. Data Warehouse Design

Kimball star schema in PostgreSQL. Dimensions include dim_customer (SCD Type 2), dim_account, dim_branch, dim_merchant, dim_card, dim_channel, dim_transaction_type, and dim_date. Facts include fact_transactions, fact_transfers, fact_atm_transactions, and fact_card_transactions. Surrogate keys join facts to dimensions.

dbt layers

- Staging: clean, cast, standardize statuses, deduplicate

- Intermediate: enriched metrics and anomaly features
- Warehouse: dimensions + incremental facts
- Analytics: business marts (daily/monthly, customer, branch, channel, failures, anomalies)

5. Data Quality & Reconciliation

Source CSVs intentionally include a small rate of duplicates, missing values, inconsistent capitalization, and invalid foreign keys. Staging standardizes these. DQ results land in `audit.data_quality_results`. Reconciliation compares distinct source business keys to warehouse counts/sums (PASS on demo run). Balance reconciliation samples accounts for opening + credits – debits vs reported closing.

6. Transaction Anomaly Detection Prototype

IMPORTANT: This is an analytical prototype, not a production fraud decision engine. Outputs are labeled "Potential anomaly" / "Transaction requiring review."

- Rule-based: high amount, high frequency, rapid window, night activity, multiple failures, sudden behavior change
- Statistical: per-customer z-score (configurable threshold, default 3.0)
- Machine learning: Isolation Forest (contamination 0.02)
- Persisted to `monitoring.transaction_anomalies` with severity Low/Medium/High/Critical

Demo anomaly counts

Severity	Count
Low	5,041
Medium	1,573
High	2,031
Total	8,645

7. Business Insights (from live SQL)

Generated from analytics marts / warehouse — synthetic data only.

Channels

Channel	Transactions	Notes
Mobile App	31,713	34.6% of top-channel volume
ATM	18,233	
POS	14,799	
Internet Banking	14,795	
Branch	12,088	

Customer segments

Inactive: >90 days since last txn (dataset as-of) or 0 txns; Low <10; Medium 10–49; High 50–199; Premium ≥200.

Segment	Customers
Medium Activity	2,471
Inactive	1,955
High Activity	388
Low Activity	186

Top branches by completed value

Branch	Region	Value	Fail %
NovaBank Dallas #50	Southwest	\$1,149,418	5.00
NovaBank Philadelphia #7	Northeast	\$1,021,449	5.48
NovaBank Detroit #47	Midwest	\$1,000,420	4.96
NovaBank Minneapolis #20	Midwest	\$955,959	4.63
NovaBank Atlanta #22	Southeast	\$894,079	4.30

8. BI Dashboard & Deployment

An interactive HTML BI dashboard is published on Vercel with executive KPIs, monthly trends, channel/region charts, customer segments, branch rankings, and anomaly review tables. Power BI Desktop specs and DAX measures are documented under dashboards/powerbi/ for native .pbix rebuilds.

- Live site: <https://banking-transaction-analytics.vercel.app>
- Dashboard: <https://banking-transaction-analytics.vercel.app/dashboard>
- GitHub: <https://github.com/alirazach476/banking-transaction-analytics>

9. Testing, Orchestration & Ops

- pytest: 17 tests covering generation, validation, anomaly logic (all passing in verified run)
- Airflow DAG banking_daily_pipeline for daily orchestration
- Docker Compose for Postgres (+ Airflow); embedded Postgres fallback for local Windows

- Makefile / python -m src.pipeline.run_pipeline for end-to-end execution
- Idempotent ingest via truncate/upsert + unique business keys; incremental watermarks supported

10. Limitations & Future Work

- Demo-scale defaults; full 2M+ transaction volumes are configurable but heavier to run
- Anomaly prototype is not a production fraud engine and has no labeled precision/recall claim
- Native Power BI .pbix is specified, not auto-generated
- Future: cloud warehouse (Redshift/Synapse/BigQuery), streaming ingest, SCD2 expansion, CI orchestration

End of report — NovaBank Banking Transaction Analytics Pipeline (synthetic data).